



Cape Peninsula
University of Technology

FACULTY OF INFORMATICS & DESIGN

Diploma in Information & Communication Technology

PROJECT MANAGEMENT 3 – PRM372S

Community Store Mobile First Marketplace

DUE DATE: 07 APRIL 2026

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Course Code	PRM372S

**Cape Peninsula University of Technology
Department of Information Technology**

Plagiarism Declaration

Subject: Project Management 3

Due Date: 07 April 2026

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- ☐ an individual project or
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PROJECT CHARTER

Project Title: Community Store Mobile-First Marketplace

1.1 Project Overview

The project is intended to design and develop a centralized digital platform that will enable the buying, selling, and exchange of products and services among the members of a particular community, specifically targeting students, faculty members, local vendors, and members of the surrounding communities. The project will be accessible via mobile devices, therefore providing ease, accessibility, and convenience for the members.

The project will be managed using the Agile project management methodology. Using the Agile methodology will enable the project team to be flexible in developing the project, thus meeting the project's requirements.

The project will result in the creation of a functional prototype that will demonstrate the key features of a marketplace.

1.2 Scope Overview

The scope of this project includes the planning, design, development, testing, and delivery of a mobile first community marketplace application. The system will incorporate essential features that support seamless interaction between users and ensure a secure and efficient trading environment.

1.2.1 In-Scope Features:

The scope of this project entails the planning, design, development, testing, and delivery of a mobile first community marketplace application that will incorporate key features that facilitate smooth interaction between users.

- User registration, login, and authentication
- User profile management
- Listing of products and services
- Advanced search and filtering functionality
- Shopping cart and checkout process
- Payment integration (simulated or functional)
- In-app messaging and notifications

- Rating and review system
- Community bulletin board for announcements and services

1.2.2 Out-of-Scope Items:

- Full-scale commercial deployment
- Advanced AI-based recommendations
- Integration with complex external enterprise systems
- Long-term maintenance beyond project duration

The project's scope is limited to delivering a realistic, functional academic solution within the given timeframe and resource constraints.

1.3 Business Case

The existing means of purchase and sale of items such as informal social media groups or other external online sources available to students and the community at large have been fragmented, untrustworthy, and lack the necessary security measures. All these factors have resulted in inefficiency, which has led to an increase in fraudulent activities.

The proposed solution for the Community Store Mobile-First Marketplace is aimed at providing a dedicated marketplace environment for the needs of the campus-based or external community at large. The solution is aimed at providing a better experience for the users of the marketplace environment.

The proposed solution is aimed at encouraging sustainability within the community since it will provide an environment where items can be reused within the same community. Moreover, the solution will provide an environment where people can get connected socially.

1.4 Background

The motivation behind this project is based on the growing need for a reliable and community-oriented digital marketplace. In most academic environments, the students and members of staff often have the need to trade or exchange goods such as books, electronics, furniture, and even services. However, the available solutions have not been sufficiently tailored to meet the needs of a community-based marketplace.

The available solutions have been too generic or lack a local trust system, which has made them ineffective for a close-knit community such as the one in the academic environment. As a result, users are forced to use unstructured and unsecured systems for their transactions.

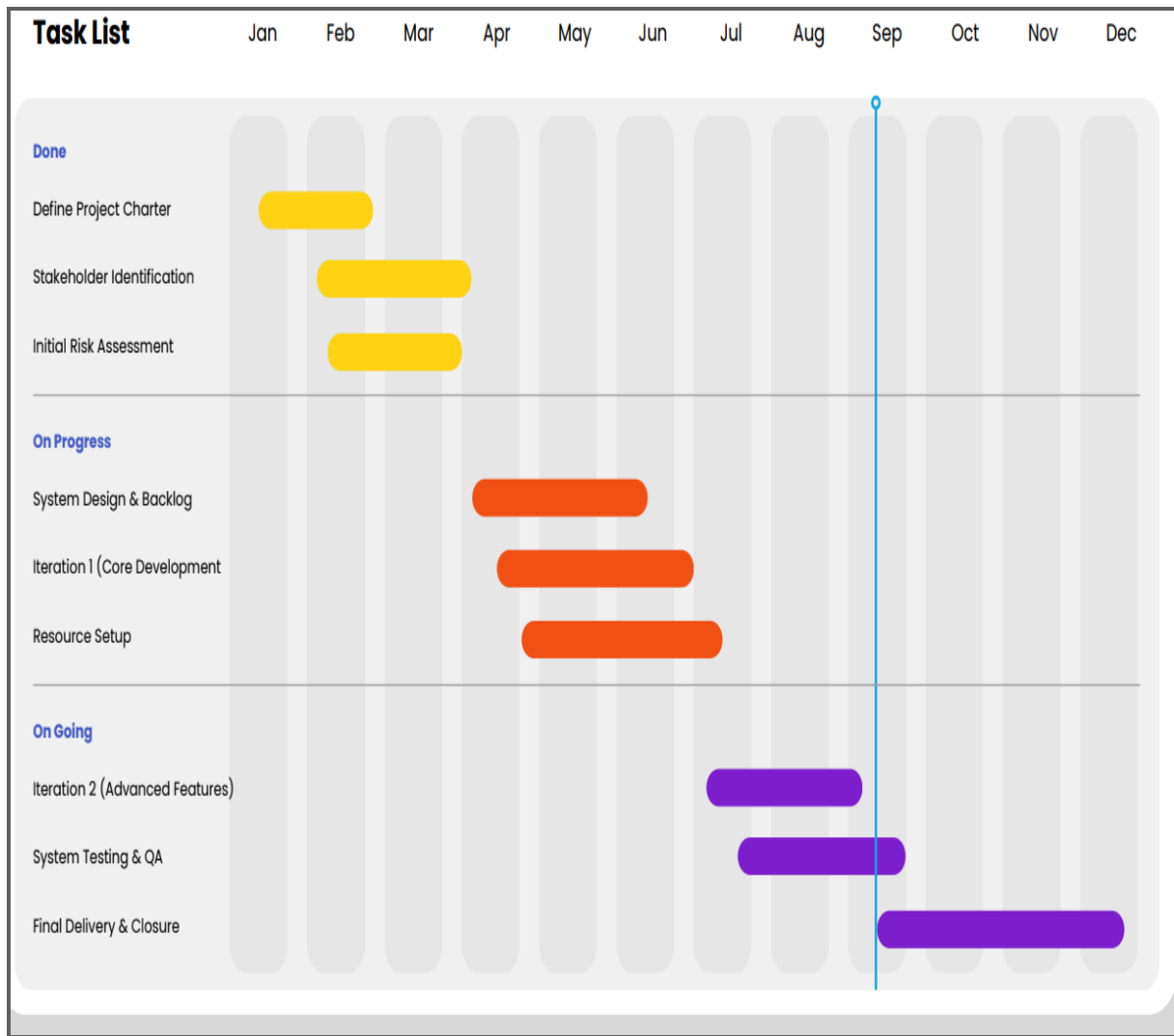
As a result, the project attempts to bridge the gap by providing a centralized solution that is specific to a university and community-based marketplace. By focusing on usability, trust, and accessibility, the project attempts to provide a reliable solution for enhancing the level of convenience.

1.5 Milestone Schedule and Acceptance Criteria

The project will be executed in structured phases aligned with Agile principles, where each milestone represents a key deliverable or checkpoint in the project lifecycle. Acceptance criteria are defined to ensure that each milestone meets the required quality standards and project objectives before progressing to the next phase.

Milestone	Target Date	Key Deliverables	Acceptance Criteria
Project Charter Approval		Finalized Project Charter	Charter reviewed and agreed upon by all group members
Requirements Elicitation & Stakeholder Analysis		Stakeholder Register, Requirements Document	All relevant stakeholders identified and requirements clearly defined and validated
System Design & Agile Backlog Development		Wireframes, System Design, Product Backlog	Design approved by team; backlog is complete and prioritised

Iteration 1 - Core Functionality Development		User Authentication, Listings Feature	Core features implemented and meet defined user stories
Iteration 2 - Advanced Features Development		Payment Simulation, Messaging, Reviews	Features functional and tested successfully
System Testing & Quality Assurance		Test Cases, Bug Reports, Improved System	Major issues resolved; system meets functional requirements
Final Delivery & Project Closure		Final System, Report, Presentation	All deliverables completed and submitted according to requirements



1.6 Risks, Assumptions, and Constraints

The successful implementation of the Community Store Mobile-First Marketplace Project hinges on the proactive identification and management of uncertainties, conditions, and project constraints. This chapter discusses the important risks that will influence the project's performance, the assumptions made in project planning, and the constraints of the project. These factors contribute to effective decision making, risk mitigation techniques, and practical project implementation.

Risks and Risk Mitigation Techniques

Risk can be defined as any event or condition that poses uncertainty and may have negative impacts on the scope, timeline, quality, or success of the project if not handled appropriately. Below are some of the risks that have been identified, together with the appropriate mitigation techniques:

Trust and Safety Risk

One of the major risks facing marketplaces is the issue of fraud, deceptive listing of products or services, or user deception. Such actions may affect users' confidence in the marketplace, and therefore adversely impact the credibility and sustainability of the project

■Mitigation Strategy:

In order to mitigate this particular risk, the system will require the users to verify themselves via strong authentication methods such as institutional emails for students and formal business registration for suppliers. Another measure will include introducing a rating and review system where the users will be encouraged to rate each other. In sum, these measures will create a trust-based environment that will discourage fraud.

Risk of Breach of Payment Security

The process of handling money on a digital platform comes with the threat of data leaks, unauthorized access, and failed payments that might cost both parties financially. Any failure in managing the transaction process efficiently would cause loss of money, harm reputation, and reduce user loyalty.

■Mitigation Strategy:

The platform will utilize reliable online payment services such as Payfast and SnapScan which guarantee the secure payment process. Moreover, it will implement additional security measures such as data encryption and two-factor authentication for large payments.

Risk of Failure to Attract Users

As the proposed solution is intended to address an informal market, there is a substantial risk that people might refuse to migrate from their informal platforms and continue using them instead of the new service.

Project Constraints

Project constraints can be described as limitations within which the project is to be undertaken. They are significant for the planning and management of projects and resource allocation.

a) Time Constraint

According to the requirements of the project, it is necessary to complete all project tasks, testing, and presentation of the results by 04 September 2026. The time constraint affects the number of features that will be integrated into the application.

b) Methodological Constraint

The main methodological limitation for this project is following the principles of the Agile methodology in project implementation. In this case, all project activities are conducted in line with the principles of this approach.

c) Resource Constraint

In accordance with the requirements, the available resources for the completion of the project will be quite limited, given that it is an academic task. Thus, the use of special software and other paid services will not be possible.

d) Scope Constraint

The scope of the project has to be realistic enough, taking into account the fact that it is an academic task. The project should emphasize the development of the application functionality and not additional features.

1.7 Resource Estimates

- **Personnel:** A team of 5 students including a Project Manager, Backend/Frontend Developers, QA Engineers, Security Specialist, and Community Liaison.
- **Technical:** Cloud hosting (AWS/GCP), microservices architecture, and Redis for caching.
- **Budget:** Minimal financial cost, leveraging student developer packs and free-tier cloud resources.

1.8 Stakeholder List

- **Internal:** Project Management Team (5 students) and Course Lecturers.
- **External:** Students - Primary buyers/sellers of textbooks and electronics.
 - **Faculty:** Users seeking or offering campus-related services.
 - **Local Vendors & Residents:** Providers of goods and services to the campus community

1.9 Team Operating Principles

- **Agile Collaboration:** Bi-weekly sprint planning, daily stand-ups (via Discord/WhatsApp) and formal sprint retrospectives.
- **Decision Making:** Technical decisions led by the Technical Planning Lead; timeline and stakeholder conflicts resolved by the Project Manager.
- **Quality Standard:** All code must undergo peer review and QA testing before being marked as "Done" in the backlog.

1.10 Lessons Learned

- **Success:** Early definition of specialized roles (Security Specialist) allowed for proactive planning against fraud.
- **Challenge:** Initial alignment on Agile tooling requires extra time; future sprints will prioritize immediate backlog entry.
- **Improvement:** Continuous monitoring of system health with real-time analytics will be integrated early to prevent performance bottlenecks.

3. STAKEHOLDER ANALYSIS AND ENGAGEMENT

Stakeholder analysis is an important part of project planning because it helps to identify all the people and groups who may affect the project or be affected by it. In this project, the Community Store Mobile-First Marketplace involves different stakeholders such as students, lecturers, vendors, and technical support teams.

The purpose of this section is to identify these stakeholders, understand their roles and interests, and evaluate their level of influence on the project. This will also help the project team decide how to communicate and work with each stakeholder effectively.

The stakeholders discussed in this section are based on those identified in the Project Charter, with additional stakeholders included to provide a more complete and detailed analysis.

analyzed based on their role, interests, level of influence, and how they will be engaged throughout the project.

Stakeholder	Role in Project	Interest	Influence (Power)	Engagement Strategy
Project Team	Team responsible for planning, development, and implementation of the system	Successful project completion, good academic performance	High	Bi-weekly meetings on platforms such as WhatsApp and Teams to allow task tracking / collab
Lectures	They provide academic guidance and evaluate deliverables	High-quality work, meeting academic requirements	High	Regular consultations, feedback sessions, formal submissions
Students	Primary users such as buyers and sellers on the platform	Affordable products, ease of use, secure transactions	High	Surveys, user testing, feedback collection
Faculty	Secondary users of the platform	Convenience and access to services	Medium	Platform updates, communication and feedback

Local vendors/Residents	Provide goods and services to the platform	Profit generation, increased visibility and customer reach	High	Onboarding processes, communication, feedback sessions
CPUT IT Department	Responsible for system integration and authentication support	Security, system compatibility, compliance with institutional standards	Very High	Formal communication, approvals, technical reporting
Payment Providers	Facilitate secure financial transactions High	Reliable, secure, and efficient payment processing	High	Technical integration, monitoring, communication

2. Stakeholder Analysis

In order to understand stakeholder involvement in the project, each stakeholder is

3. Power-Interest Analysis

After identifying the stakeholders, it is important to understand their level of power and interest in the project. This helps the project team to decide how much attention should be given to each stakeholder.

Stakeholders with both high power and high interest need to be managed closely because they have a strong influence on the success of the project. In this case, the project team, lecturers, and the CPUT IT Department fall into this category, as they are directly involved in decision-making and project approval.

On the other hand, stakeholders with high power but lower interest, such as payment providers, need to be kept satisfied. Although they may not be involved in daily project activities, their role is still important for the system to function properly.

Stakeholders with high interest but lower power, such as students and local vendors, should be kept informed. These users are important because they will be using the system, and their feedback can help improve the platform.

Lastly, stakeholders with low power and low interest, such as the broader community, should be monitored. While they are not directly involved, they may still be affected by the project.

4. Stakeholder Engagement Strategy

Effective communication with stakeholders is necessary to ensure the success of the project. Different strategies will be used depending on the type of stakeholder.

For internal stakeholders such as the project team, communication will take place through regular meetings, including bi-weekly discussions on platforms such as WhatsApp and Microsoft Teams. This will help to track progress and ensure that all members are aligned with project goals.

For external stakeholders such as students, faculty, and local vendors, engagement will take place through feedback collection, onboarding processes, and system updates. This ensures that their needs are understood and considered during development.

In addition, high-power stakeholders such as lecturers and the CPUT IT Department will be engaged through formal communication methods, including progress reports and consultations. This is important to ensure that the project meets academic and technical requirements.

3. Requirements Document

1. Functional Requirements (FR)

what the system must do to satisfy stakeholder needs.

- FR1: User Authentication (CPUT IT Dept. / Students): The system shall restrict registration to users with a valid university email address (cput.ac.za) to ensure a trusted environment.
- FR2: Product & Service Listings (Students / Vendors): The system shall allow authenticated users to create, edit, and delete listings for physical goods (e.g., textbooks) and services (e.g., tutoring).
- FR3: Advanced Search & Filtering (Students): Users shall be able to filter listings by category, price range, and item condition (New/Used).
- FR4: Secure Payment Processing (Payment Providers): The system shall integrate with PayFast or SnapScan to facilitate secure financial transactions between parties.
- FR5: Community Bulletin Board (Faculty / Residents): The platform shall include a dedicated section for non-commercial community announcements and local event promotions.
- FR6: Trust-Based Rating System (All Users): The system shall enable a five-star rating and text review feature for every completed transaction to build seller credibility.

2. Non-Functional Requirements (NFR)

how the system must perform to meet quality standards.

- NFR1: Security (Security Specialist): All sensitive data, including login credentials and payment tokens, must be encrypted using industry-standard protocols (HTTPS/SSL).
- NFR2: Performance & Scalability: The system shall utilize a microservices architecture and Redis caching to ensure search results load in under 2 seconds during peak campus hours.
- NFR3: Mobile-First Usability: The user interface must be fully responsive and optimized for mobile devices, ensuring accessibility for students on the move.

3. Project Constraints & Assumptions

define the boundaries of your academic project.

- Constraint 1: Timeline: All core functionality and final documentation must be completed by the hard deadline of September 4, 2026.
- Constraint 2: Budget: As an academic task, the project must leverage free-tier cloud resources and student developer packs; no paid software services can be used.
- Assumption 1: It is assumed that the CPUT IT Department will provide the necessary API access or support for institutional email verification.

4. INITIAL RISK REGISTER

Risk ID	Risk Description	Likelihood	Impact	Mitigation Strategy	Owner
R1	Fraudulent Listings: Users posting fake products or deceptive descriptions .	Medium	High	Implement mandatory institutional email verification and a flagging system.	Community Liaison
R2	Payment Security Breach: Unauthorized access to payment data or failed transactions .	Low	High	Integrate trusted gateways (PayFast/SnapScan) and use two-factor authentication.	Security Specialist
R3	Low User Adoption: Students prefer existing	High	Medium	Launch awareness campaigns and offer incentives	Project Manager

	tools like WhatsApp or Facebook.			like free listings	
R4	Timeline Delays: Failure to meet the Sept 4th deadline due to scope creep.	Medium	High	Use Agile bi-weekly sprint reviews to strictly monitor progress and adjust scope.	Project Manager